Tutorial 7 Solution

Problem 1

Show that if $g : \mathbb{R}^2 \to \mathbb{R}$ is a convex function and $b \in \mathbb{R}$ then $S = \{(x, y) \in \mathbb{R}^2 : g(x, y) \leq b\}$ is a convex set.

Solution: We have to show that if $(x, y) \in S, (x', y') \in S$ and $\lambda \in [0, 1]$, then $\lambda(x, y) + (1 - \lambda)(x', y') \in S$. So let $(x, y) \in S, (x', y') \in S$ and $\lambda \in [0, 1]$.

- $(x, y) \in S$ implies $g(x, y) \leq b$ and thus

$$\lambda g(x, y) \leq \lambda b \tag{1}$$

- $(x', y') \in S$ implies $g(x', y') \leq b$ and thus

$$(1 - \lambda)g(x', y') \leq (1 - \lambda)b \tag{2}$$

Thus

$$\begin{aligned} g(\lambda(x, y) + (1 - \lambda)(x', y')) &\leq \lambda g(x, y) + (1 - \lambda)g(x', y') \\ &\leq \lambda b + (1 - \lambda)b = b, \end{aligned}$$

where the first inequality follows from the convexity of g and the second inequality from (1) and (2). But $g(\lambda(x, y) + (1 - \lambda)(x', y')) \leq b$ means that $\lambda(x, y) + (1 - \lambda)(x', y') \in S$. This is what we wanted to show. Thus S is convex.

Note. $S = \{(x, y) \in \mathbb{R}^2 : g(x, y) \leq b\}$ is called the **lower contour set** of g for level b. Similarly, $S' = \{(x, y) \in \mathbb{R}^2 : h(x, y) \geq a\}$, which is called the **upper contour set** of h for level a. If $h : \mathbb{R}^2 \to \mathbb{R}$ is a concave function and $a \in \mathbb{R}$ then S' is a convex set for every $a \in \mathbb{R}$.

Problem 2

Suppose $f : \mathbb{R}^2 \to \mathbb{R}$ can be written as

$$f(x, y) = g(x) + h(y),$$

where both g and h are convex functions of one variable. Show that f is also convex.

Solution: To show that f is convex, take any $(x', y'), (x'', y'') \in \mathbb{R}^2$ and $\lambda \in [0, 1]$. Now

$$\begin{aligned} f(\lambda(x', y') + (1 - \lambda)(x'', y'')) &= f(\lambda x' + (1 - \lambda)x'', \lambda y' + (1 - \lambda)y'') \\ &= g(\lambda x' + (1 - \lambda)x'') + h(\lambda y' + (1 - \lambda)y''). \end{aligned}$$

Because both g and h are convex,

$$g(\lambda x' + (1 - \lambda)x'') \leq \lambda g(x') + (1 - \lambda)g(x''),$$

and

$$h(\lambda y' + (1-\lambda)y'') \leq \lambda h(y') + (1-\lambda)h(y'').$$

Thus,

$$\begin{aligned} f(\lambda(x',y') + (1-\lambda)(x'',y'')) &= g(\lambda x' + (1-\lambda)x'') + h(\lambda y' + (1-\lambda)y'') \\ &\leq \lambda g(x') + (1-\lambda)g(x'') + \lambda h(y') + (1-\lambda)h(y'') \\ &= \lambda[g(x') + h(y')] + (1-\lambda)[g(x'') + h(y'')] \\ &= \lambda f(x',y') + (1-\lambda)f(x'',y'') \end{aligned}$$

By definition, $f(x,y)$ is convex.

Problem 3

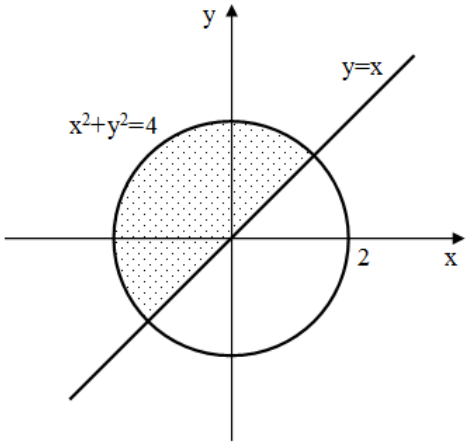
Show that the following sets are convex.

1. $S = \{(x,y) \in \mathbb{R}^2 : y \geq x,\ x^2 + y^2 \leq 4\}$
2. $T = \{(x,y) \in \mathbb{R}^2 : y \geq x^2,\ y \leq 4\}$

Solution:

1. We observe that
   - $S_1 = \{(x,y) \in \mathbb{R}^2 : y \geq x\}$ is convex because $S_1 = \text{epi}(f_1)$, where $f_1 : \mathbb{R} \to \mathbb{R}$ is given by $f_1(x) = x$ is a convex function (because it's linear, and hence affine; affine functions are both convex and concave).
     (Alternatively, one may note that $S_1 = \{(x,y) \in \mathbb{R}^2 : x - y \leq 0\}$. One can use the result we proved in Problem 2 to show that $f_a(x,y) = x - y$ is a convex function (let $g(x) = x$ and $h(y) = -y$) and then apply the result we proved in Problem 1.)
   - $S_2 = \{(x,y) \in \mathbb{R}^2 : x^2 + y^2 \leq 4\}$ is convex by the result in Problem 1 because we proved in lecture 7 that the function $f_2(x,y) = x^2 + y^2$ is convex.

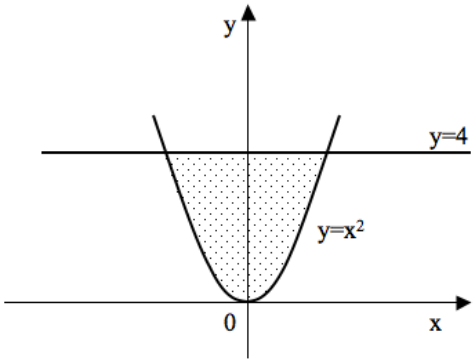
   Thus, since $S = S_1 \cap S_2$ and since the intersection of convex sets is convex, S is convex. Graphically,



2. We observe that

   - $T_1 = \{(x, y) \in \mathbb{R}^2 : y \geq x^2\}$ is convex because $T_1 = \text{epi}(g_1)$, where $g_1(x) = x^2$ is a convex function (one can prove it by taking second derivative to get $g_1''(x) = 2 > 0$, or prove it using definition as in Problem 5).
     (Alternatively, one may note that $T_1 = \{(x, y) \in \mathbb{R}^2 : x^2 - y \leq 0\}$. One can use the result we proved in Problem 2 to show that $g_a(x, y) = x^2 - y$ is a convex function (let $g(x) = x^2$ and $h(y) = -y$), and then apply the result in Problem 1.)
   - $T_2 = \{(x, y) \in \mathbb{R}^2 : y \leq 4\}$ is convex by Problem 1 because $g_2(x, y) = y$ is a convex function by the result we proved in Problem 2.

   Thus, since $T = T_1 \cap T_2$ and since the intersection of convex sets is convex, T is convex. Graphically,



Problem 4

Analyze whether the following functions are concave or convex.

1. $f_1(x, y) = 4x^2 + e^{2y^2}$.
2. $f_2(x, y) = \ln(3x + 2y)$.

Solution:

1. Observe that $f_1(x, y) = g_1(x) + h_1(y)$, where $g_1(x) = 4x^2$, and $h_1(y) = e^{2y^2}$. Now $g_1(x)$ is convex, and $h_1(y) = m \circ n$, where $m(u) = e^u$ is increasing and convex, and $n(y) = 2y^2$ is convex. Thus, both g_1 and h_1 are convex; by the result we proved in Problem 2, $f_1(x, y)$ is also convex.
2. $f_2(x, y) = \ln(3x + 2y)$ is concave because $f_2 = h_2 \circ g_2$, where $h_2(u) = \ln u$ is an increasing concave function, and $g_2(x, y) = 3x + 2y$ is linear and hence concave.

Problem 5

Prove from the definition of a strictly convex function and without using any theorem about convex functions that $f(x) = x^2$ is strictly convex.

Solution: f is strictly convex if for all $x', x'' \in \mathbb{R}$ such that $x' \neq x''$, and all $\lambda \in (0,1)$,

$$f(\lambda x' + (1-\lambda)x'') < \lambda f(x') + (1-\lambda) f(x'').$$

So let $x', x'' \in \mathbb{R}$ be such that $x' \neq x''$, and $\lambda \in (0,1)$. Then

$$f(\lambda x' + (1-\lambda)x'') = (\lambda x' + (1-\lambda)x'')^2 = \lambda^2 (x')^2 + 2\lambda(1-\lambda)x'x'' + (1-\lambda)^2 (x'')^2$$

and

$$\lambda f(x') + (1-\lambda) f(x'') = \lambda (x')^2 + (1-\lambda)(x'')^2.$$

Now observe that

$$\begin{aligned}
&\lambda f(x') + (1-\lambda) f(x'') - f(\lambda x' + (1-\lambda)x'') \\
&= \lambda (x')^2 + (1-\lambda)(x'')^2 - \left[\lambda^2 (x')^2 + 2\lambda(1-\lambda)x'x'' + (1-\lambda)^2 (x'')^2\right] \\
&= (\lambda - \lambda^2)(x')^2 - 2\lambda(1-\lambda)x'x'' + \left[1 - \lambda - (1-\lambda)^2\right](x'')^2 \\
&= \lambda(1-\lambda)(x')^2 - 2\lambda(1-\lambda)x'x'' + (1-\lambda)[1-(1-\lambda)](x'')^2 \\
&= \lambda(1-\lambda)\left[(x')^2 - 2x'x'' + (x'')^2\right] \\
&= \lambda(1-\lambda)(x' - x'')^2 > 0
\end{aligned}$$

where the strict inequality holds since $x' \neq x''$ and $\lambda \in (0,1)$. Hence $f(\lambda x' + (1-\lambda)x'') < \lambda f(x') + (1-\lambda) f(x'')$ and f is strictly convex.

Problem 6 (Extra Practice Problem)

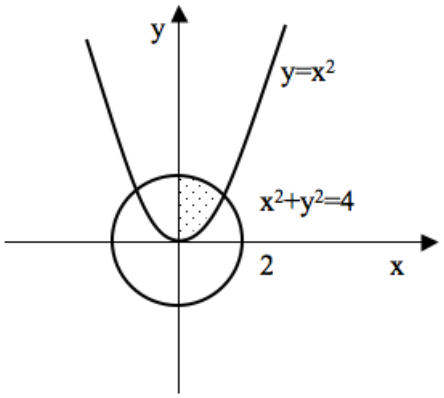
Apply the result from Problem 1 and the properties of a convex set to demonstrate that the following sets are convex.

1. $S = \{(x,y) \in \mathbb{R}^2 : y \geq x^2,\ x^2 + y^2 \leq 4,\ x \geq 0,\ y \geq 0\}$
2. $T = \{(x,y) \in \mathbb{R}^2 : x^2 + y^2 \leq 25,\ x + y \geq 5,\ y \geq 2,\ x \geq 0\}$

Solution: (Sketch only)

1. S is convex, as it is the intersection of the following sets, all of which are convex:
   - $S_1 = \{(x,y) \in \mathbb{R}^2 : y \geq x^2\}$ (we showed in Problem 3 that it is convex),
   - $S_2 = \{(x,y) \in \mathbb{R}^2 : y^2 + x^2 \leq 4\}$ (we showed in Problem 3 that it is convex),
   - $S_3 = \{(x,y) \in \mathbb{R}^2 : -x \leq 0\}$ (convex by Problem 1 because $f_3(x,y) = -x$ is affine and thus convex)
   - $S_4 = \{(x,y) \in \mathbb{R}^2 : -y \leq 0\}$ (convex by Problem 1 because $f_4(x,y) = -y$ is affine and thus convex)

   Graphically:



2. T is convex, as it is the intersection of the following sets, all of which are convex:
   - $T_1 = \{(x, y) \in \mathbb{R}^2 : y^2 + x^2 \leq 25\}$ (convex by Problem 1 because $g_1(x, y) = x^2 + y^2$ is convex, as we proved in lecture 7),
   - $T_2 = \{(x, y) \in \mathbb{R}^2 : y \geq 5 - x\}$ (convex because $T_2 = \text{epi}(g_2)$, where $g_2(x) = 5 - x$ is affine and thus convex),
   - $T_3 = \{(x, y) \in \mathbb{R}^2 : -x \leq 0\}$ (convex by Problem 1 because $f_3(x, y) = -x$ is affine and thus convex)
   - $T_4 = \{(x, y) \in \mathbb{R}^2 : -y \leq -2\}$ (convex by Problem 1 because $f_4(x, y) = -y$ is affine and thus convex)

   Graphically:

